

Readmission Rate by Pollutants

Tableau Dashboard , Reporting & Visualization

Dashboard Overview

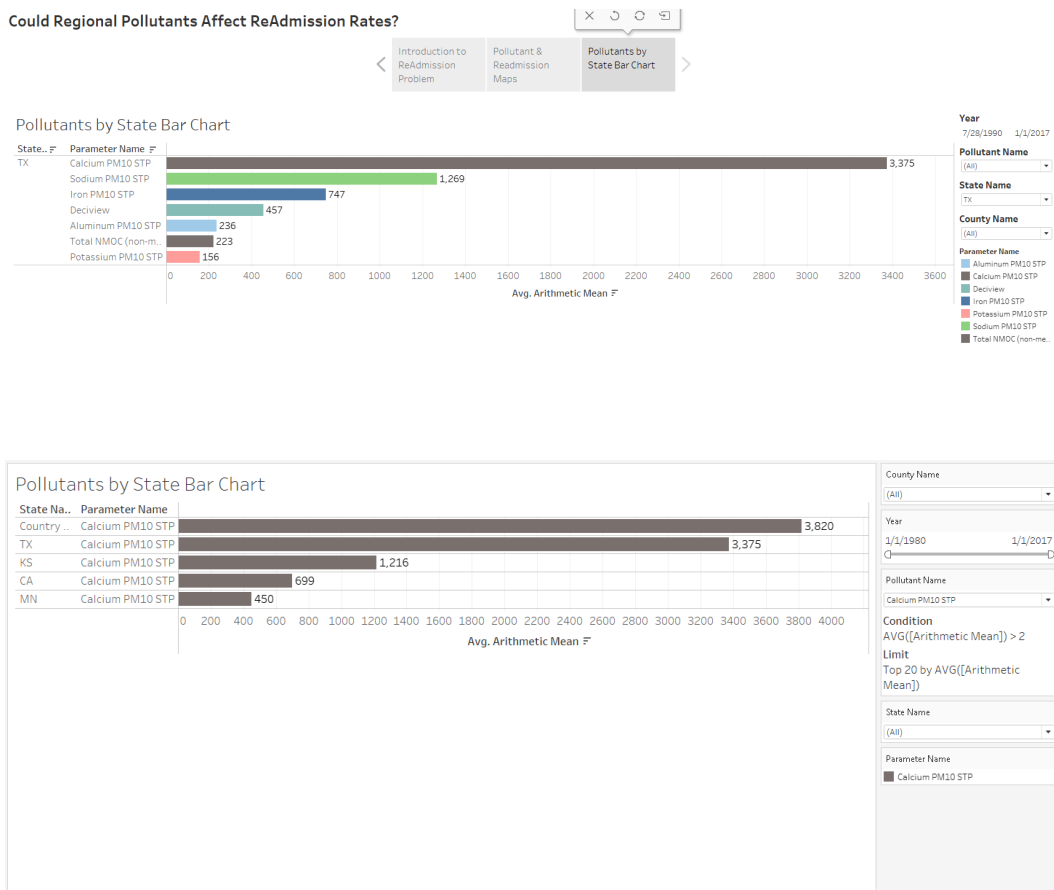
The dashboard integrates data from two sources: the provided medical dataset and an external EPA air quality annual summary public dataset. It includes four different data representations to summarize the data and display trends, two interactive controls that allow the user to modify the presentation of data, and two different metrics and key performance indicators computed using data from both chosen data sets.

Data Sources

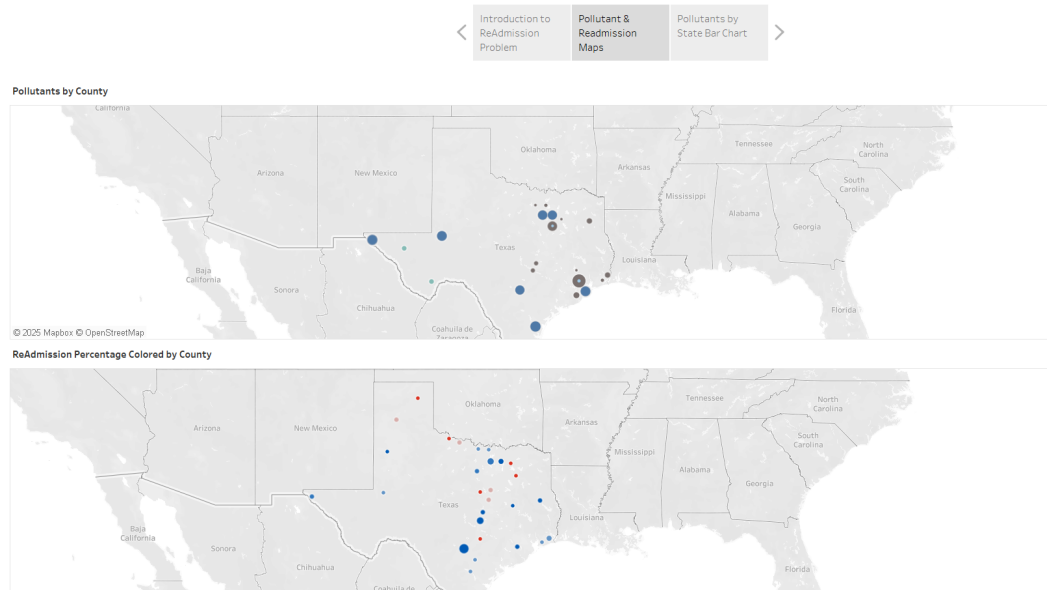
medical_clean.csv and <https://www.kaggle.com/datasets/epa/air-quality>

Key Results

It is clear patients who spend more time in the hospital shown by the initial days measurement are at higher risk of readmission. Presumably, these patients are often difficult to diagnose and treat, as they need the most time in the hospital and return at the highest rate despite receiving the most treatment.



Could Regional Pollutants Affect ReAdmission Rates?



Actionable Insights

Hospitals with high readmission rates should perform an analysis using the dashboard controls to determine the greatest pollutants in their state/county and then research if those pollutants have been linked to health issues. For example, a hospital in Texas may benefit from testing patients for the pollutant Calcium PM10 STP. This is not only the largest pollutant by volume in the state of Texas, but Texas is outputting more than twice as much of this pollutant than any other state. Total hydrocarbons and methane emissions could also be further researched and tested for correlation with chronic health issues that cannot otherwise be attributed and are potentially causing readmission.

Dashboard Purpose & Alignment

The medical data dictionary states that hospitals are struggling with patient re-admission rates being too high. As hospitals are unaware of what factors could be causing this, there may be useful insights in finding correlations between pollutants and nearby hospital re-admission rates. This dashboard will allow end users to build a map visualizing the highest hospital re-admission rates and volume, allowing them to visualize which pollutants are most common nearby. The pollutant by state bar chart allows deeper analysis, showing the 20 highest pollutants by arithmetic mean, with the option to group and filter by state or county. These data visualizations are meant to suggest an alternative perspective into potential chemical exposure/inhalation causing otherwise undiagnosable and untreatable readmissions.

Additional Dataset Value

The pollutant data set will allow visualizations using state and county, which can be the linking field between these two data fields. This will allow for a worksheet showing highest pollutants by region to compare to a map of pollutants. The pollutant dataset includes measures of each pollutant, pollutant names, measurement units, location of measurement, and other details are included to allow for detailed geographic

correlation and annual analysis.

Data Representations for Decision-Making

The county dashboard can be used by executives to assess the counties with the highest readmission rates in the country, filtering out counties with lower readmission rates for clarity. This can be compared to the map directly above, which shows the geographic distribution of pollutants. This allows a filter on parameter name to show however many pollutants at a time are desired. Finding the pollutants that stand out as having a noticeably higher distribution in the areas with the highest readmission rates could indicate that hospitals should begin testing for pollutant buildup in area patients.

The pollutants by state bar chart worksheet is set up for executives to analyze pollutant data by either selecting the pollutants they would like to see distributed across states, or select state/county names to analyze which pollutants are most prevalent in a given area. This will populate a list of pollutants in chosen states and counties to suggest which pollutants to research or keep data on.

Interactive Controls

The pollutant measurements dashboard has both a filter for pollutant name and state name. This allows users to visualize which pollutants are most prevalent per state, and the ability to visualize which states include any selected pollutants.

The county dashboard includes a sliding filter to include readmission rates only within any chosen range which will visualize on the map immediately.

Storytelling Elements

The data has been presented to show executives that pollutants have been significantly rising over time using the annual pollution bar chart, alongside the problematic issue of readmission rates being so high for patients with longer initial stays using the readmission rate by initial days visualization. We will visualize and discuss how these pollutants have grown without an understanding of future health detriments, and how these factors go largely ignored in health analyses.

Then, a more local area such as the state of Texas can be chosen. Readmission rates across the state can be shown by color on the pollution by county map, and then pollutants in the state can be listed using the pollutants by state bar chart.

Strong visualization is the most important element of storytelling in this presentation. Each map or chart has been carefully colored to ensure clear distinction between points while drawing in attention. Filters have been edited to be as clear and concise as possible while allowing users full customization. All text shown in the presentation has been minimized to keep the focus on the data, but necessary clarifications have been input.

An element this presentation relies on is the uncertainty around readmission rates and these pollutants, that can cause a bit of discomfort to hook the audience. Showing the audience that we don't understand why

patients are being readmitted and following it up with discussion of how little these measured pollutants are understood introduces a significant element of unknown that our brains very much want to resolve to break that sense of incompleteness. This presentation is framed as a potential answer to some health issues, it suggests a correlation while allowing audience members to conclude if they find these pollutants a potential factor in readmission or not, encouraging the audience to explore the visualizations more deeply using the readily available filters.

Audience Analysis

The medical data dictionary outlines that this presentation will be shown to the Sr Vice President of Hospital operations, the VP of research, regional VPs, and data analytics peers. The executives are likely not well versed in building data visualizations, so plain language is used without explaining the complexity of the Tableau calculations. Presumably, the executives would like an efficient presentation that communicates a clear point without spending time on how the dashboard was built. That is why all relevant filters are already added and have clarified, bolded titles with simple dropdowns, these are fully self-explanatory while capturing all use cases. Tooltip has also been clarified to give clear details such as unit of measure and exact measurements along with location data, while removing any information that is non-necessary. Additionally, the charts and maps have been created using a wide color range to distinguish pollutants and draw the audience in with some intriguing visuals.

Universal Access

The story and relating dashboards have been optimized for end users with no tableau experience or understanding of these datasets. Data fields have been renamed to clarify and simplify what each graph or map is visualizing, and filters have also been given clarified titles and clear legend visualizations so that users with no background knowledge will be able to intuitively visualize desired data from these sets. Where appropriate, color schemes have also been added as an additional layer of visualization with legends on the right pane. All irrelevant filters, fields, or text have been removed from the worksheets, dashboards, and story. Colors have been modified for accessibility by colorblind individuals as well.